POST-PEX STA

Run the STA script:

```
cd /opt/asic-learning/countermod10

cat > physical/post_route_sta.tcl <<'TCL'
puts "\n===== LOAD TIMING LIBRARY ====="
read_liberty /sky130hd/lib/sky130_fd_sc_hd__tt_025C_1v80.lib

puts "\n===== LOAD DETAILED-ROUTED DATABASE ====="
read_db /work/physical/results/detailed_route.odb

puts "\n===== LOAD TIMING CONSTRAINTS ====="
read_sdc /work/constraints/constraint.sdc

puts "\n===== LOAD ROUTED PARASITICS ====="
read_spef /work/physical/results/post_route.spef

puts "\n===== TIMING ANALYSIS ====="
set_propagated_clock [all_clocks]

puts "\n===== SETUP TIMING ====="
report_checks -path_delay max -format full_clock_expanded -fields {slew cap fanout
input_pin nets} -digits 4

puts "\n===== HOLD TIMING ====="
report_checks -path_delay min -format full_clock_expanded -fields {slew cap fanout
input_pin nets} -digits 4

puts "\n===== SETUP WNS AND TNS ====="
report_worst_slack -max
report_tns

puts "\n===== HOLD WNS AND TNS ====="
report_worst_slack -min
report_tns -hold

puts "\n===== POST-ROUTE STA COMPLETED ====="
TCL

cd /opt/asic-learning/countermod10
```

```
sed -i \
  's/{slew cap fanout input_pin nets}/{slew cap fanout input_pin}/g; s/report_tns
-hold/report_tns -min/' \
  physical/post_route_sta.tcl
```

Run the postPEX STA:

```
sudo docker run --rm \
  -v /opt/asic-learning/countermod10:/work \
  -v /opt/asic-learning/OpenROAD-flow-scripts/flow/platforms/sky130hd:/sky130hd:ro
\
  -v /opt/asic-learning/OpenROAD-flow-scripts/flow/platforms/sky130hs:/sky130hs:ro
\
  --entrypoint /OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad \
  openroad/orfs:latest \
  -exit /work/physical/post_route_sta.tcl \
  2>&1 | tee physical/logs/post_route_sta.log
```

Post-route timing results so far

Metric	Value
Clock period	10 ns
Clock frequency	100 MHz
Setup WNS	+8.33 ns
Setup TNS	0.00 ns
Hold WNS	+0.48 ns

Worst setup path shown	Reset recovery, +9.4383 ns
Worst register-to-register setup path not yet isolated	Not confirmed
Worst hold path shown	Reset removal, +0.4782 ns
Register-to-register hold slack shown	+0.5370 ns

STA results:

```
sphoorti@sphoorti-inspiron153530:/opt/asic-learning/countermod
10$ cd /opt/asic-learning/countermod10

sed -i \
 's/{slew cap fanout input_pin nets}/{slew cap fanout
input_pin}/g; s/report_tns -hold/report_tns -min/' \
 physical/post_route_sta.tcl
sphoorti@sphoorti-inspiron153530:/opt/asic-learning/countermod
10$ sudo docker run --rm \
 -v /opt/asic-learning/countermod10:/work \
 -v
/opt/asic-learning/OpenROAD-flow-scripts/flow/platforms/sky130
hd:/sky130hd:ro \
 -v
/opt/asic-learning/OpenROAD-flow-scripts/flow/platforms/sky130
hs:/sky130hs:ro \
 --entrypoint
/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad \
 openroad/orfs:latest \
 -exit /work/physical/post_route_sta.tcl \
 2>&1 | tee physical/logs/post_route_sta.log
OpenROAD unknown
Features included (+) or not (-): -GPU +GUI -Python
This program is licensed under the BSD-3 license. See the
LICENSE file for details.
```

Components of this program may be licensed under more
restrictive licenses which must be honored.

===== LOAD TIMING LIBRARY =====

===== LOAD DETAILED-ROUTED DATABASE =====

===== LOAD TIMING CONSTRAINTS =====

===== LOAD ROUTED PARASITICS =====

===== TIMING ANALYSIS =====

===== SETUP TIMING =====
Startpoint: rst (input port clocked by core_clock)
Endpoint: _26_ (recovery check against rising-edge clock
core_clock)
Path Group: asynchronous
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
			0.0000	0.0000	clock core_clock (rise edge)
			0.0000	0.0000	clock network delay (propagated)
			1.0000	1.0000 v	input external delay
4	0.0159	0.0000	0.0000	1.0000 v	rst (in)
		0.0002	0.0001	1.0001 v	_22_/A (sky130_fd_sc_hd__clkinv_1)
1	0.0039	0.0283	0.0272	1.0273 ^	_22_/Y (sky130_fd_sc_hd__clkinv_1)
		0.0283	0.0001	1.0274 ^	_26_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
				1.0274	data arrival time
			10.0000	10.0000	clock core_clock (rise edge)
			0.0000	10.0000	clock source latency
1	0.0037	0.0000	0.0000	10.0000 ^	clk (in)

```
                   0.0001    0.0000   10.0000 ^ clkbuf_0_clk/A
(sky130_fd_sc_hd__clkbuf_1)
    2    0.0081    0.1026    0.1068   10.1068 ^ clkbuf_0_clk/X
(sky130_fd_sc_hd__clkbuf_1)
                   0.1026    0.0002   10.1070 ^
clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_1)
    2    0.0052    0.0712    0.1159   10.2229 ^
clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_1)
                   0.0712    0.0001   10.2230 ^ _26_/CLK
(sky130_fd_sc_hd__dfrtp_1)
                             0.0000   10.2230   clock
reconvergence pessimism
                             0.2427   10.4657   library
recovery time
                                      10.4657   data required
time
-----------------------------------------------------------------
------------------------
                                      10.4657   data required
time
                                      -1.0274   data arrival
time
-----------------------------------------------------------------
------------------------
                                       9.4383   slack (MET)


Startpoint: _23_ (rising edge-triggered flip-flop clocked by
core_clock)
Endpoint: count[0] (output port clocked by core_clock)
Path Group: core_clock
Path Type: max

Fanout       Cap      Slew     Delay      Time   Description
-----------------------------------------------------------------
------------------------
                             0.0000    0.0000   clock
core_clock (rise edge)
                             0.0000    0.0000   clock source
latency
    1    0.0037    0.0000    0.0000    0.0000 ^ clk (in)
                   0.0001    0.0000    0.0000 ^ clkbuf_0_clk/A
(sky130_fd_sc_hd__clkbuf_1)
```

```
    2    0.0081    0.1026    0.1068    0.1068 ^ clkbuf_0_clk/X
(sky130_fd_sc_hd__clkbuf_1)
                   0.1026    0.0002    0.1070 ^
clkbuf_1_1__f_clk/A (sky130_fd_sc_hd__clkbuf_1)
    2    0.0052    0.0712    0.1159    0.2229 ^
clkbuf_1_1__f_clk/X (sky130_fd_sc_hd__clkbuf_1)
                   0.0712    0.0001    0.2230 ^ _23_/CLK
(sky130_fd_sc_hd__dfrtp_1)
    7    0.0202    0.1157    0.4422    0.6652 v _23_/Q
(sky130_fd_sc_hd__dfrtp_1)
                   0.1157    0.0007    0.6659 v count[0] (out)
                                       0.6659   data arrival
time

                              10.0000   10.0000   clock
core_clock (rise edge)
                               0.0000   10.0000   clock network
delay (propagated)
                               0.0000   10.0000   clock
reconvergence pessimism
                              -1.0000    9.0000   output
external delay
                                        9.0000   data required
time
-----------------------------------------------------------------
------------------------
                                        9.0000   data required
time
                                       -0.6659   data arrival
time
-----------------------------------------------------------------
------------------------
                                        8.3341   slack (MET)



===== HOLD TIMING =====
Startpoint: rst (input port clocked by core_clock)
Endpoint: _24_ (removal check against rising-edge clock
core_clock)
Path Group: asynchronous
Path Type: min

Fanout       Cap      Slew     Delay      Time   Description
```

```
-------------------------------------------------------------
-----------------------
                           0.0000    0.0000   clock
core_clock (rise edge)
                           0.0000    0.0000   clock network
delay (propagated)
                           1.0000    1.0000 v input external
delay
    4    0.0159    0.0000    0.0000    1.0000 v rst (in)
                   0.0003    0.0001    1.0001 v _10_/A
(sky130_fd_sc_hd__clkinv_1)
    1    0.0039    0.0281    0.0271    1.0272 ^ _10_/Y
(sky130_fd_sc_hd__clkinv_1)
                   0.0281    0.0001    1.0273 ^ _24_/RESET_B
(sky130_fd_sc_hd__dfrtp_1)
                                       1.0273   data arrival
time

                           0.0000    0.0000   clock
core_clock (rise edge)
                           0.0000    0.0000   clock source
latency
    1    0.0037    0.0000    0.0000    0.0000 ^ clk (in)
                   0.0001    0.0000    0.0000 ^ clkbuf_0_clk/A
(sky130_fd_sc_hd__clkbuf_1)
    2    0.0081    0.1026    0.1068    0.1068 ^ clkbuf_0_clk/X
(sky130_fd_sc_hd__clkbuf_1)
                   0.1026    0.0002    0.1071 ^
clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_1)
    2    0.0055    0.0745    0.1183    0.2253 ^
clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_1)
                   0.0745    0.0001    0.2255 ^ _24_/CLK
(sky130_fd_sc_hd__dfrtp_1)
                           0.0000    0.2255   clock
reconvergence pessimism
                           0.3237    0.5491   library
removal time
                                       0.5491   data required
time
-------------------------------------------------------------
-----------------------
                                       0.5491   data required
time
```

```
                                        -1.0273   data arrival
time
-------------------------------------------------------------
-----------------------
                                         0.4782   slack (MET)


Startpoint: _25_ (rising edge-triggered flip-flop clocked by
core_clock)
Endpoint: _25_ (rising edge-triggered flip-flop clocked by
core_clock)
Path Group: core_clock
Path Type: min

Fanout       Cap      Slew     Delay      Time   Description
-------------------------------------------------------------
-----------------------
                              0.0000    0.0000   clock
core_clock (rise edge)
                              0.0000    0.0000   clock source
latency
    1    0.0037    0.0000    0.0000    0.0000 ^ clk (in)
                   0.0001    0.0000    0.0000 ^ clkbuf_0_clk/A
(sky130_fd_sc_hd__clkbuf_1)
    2    0.0081    0.1026    0.1068    0.1068 ^ clkbuf_0_clk/X
(sky130_fd_sc_hd__clkbuf_1)
                   0.1026    0.0002    0.1071 ^
clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_1)
    2    0.0055    0.0745    0.1183    0.2253 ^
clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_1)
                   0.0745    0.0001    0.2255 ^ _25_/CLK
(sky130_fd_sc_hd__dfrtp_1)
    5    0.0165    0.1535    0.4132    0.6387 ^ _25_/Q
(sky130_fd_sc_hd__dfrtp_1)
                   0.1535    0.0004    0.6391 ^ _17_/A
(sky130_fd_sc_hd__xnor2_1)
    1    0.0026    0.0327    0.0892    0.7283 v _17_/Y
(sky130_fd_sc_hd__xnor2_1)
                   0.0327    0.0001    0.7284 v _25_/D
(sky130_fd_sc_hd__dfrtp_1)
                                        0.7284   data arrival
time
```

```
                           0.0000    0.0000   clock
core_clock (rise edge)
                           0.0000    0.0000   clock source
latency
    1    0.0037    0.0000    0.0000    0.0000 ^ clk (in)
                   0.0001    0.0000    0.0000 ^ clkbuf_0_clk/A
(sky130_fd_sc_hd__clkbuf_1)
    2    0.0081    0.1026    0.1068    0.1068 ^ clkbuf_0_clk/X
(sky130_fd_sc_hd__clkbuf_1)
                   0.1026    0.0002    0.1071 ^
clkbuf_1_0__f_clk/A (sky130_fd_sc_hd__clkbuf_1)
    2    0.0055    0.0745    0.1183    0.2253 ^
clkbuf_1_0__f_clk/X (sky130_fd_sc_hd__clkbuf_1)
                   0.0745    0.0001    0.2255 ^ _25_/CLK
(sky130_fd_sc_hd__dfrtp_1)
                           0.0000    0.2255   clock
reconvergence pessimism
                          -0.0340    0.1914   library hold
time
                                     0.1914   data required
time
-----------------------------------------------------------------
-----------------------
                                     0.1914   data required
time
                                    -0.7284   data arrival
time
-----------------------------------------------------------------
-----------------------
                                     0.5370   slack (MET)



===== SETUP WNS AND TNS =====
worst slack max 8.33
tns max 0.00

===== HOLD WNS AND TNS =====
worst slack min 0.48
tns min 0.00

===== POST-ROUTE STA COMPLETED =====
sphoorti@sphoorti-inspiron153530:/opt/asic-learning/countermod
10$
```

```
===== DETAILED ROUTE DRC REPORT =====
===== POST-ROUTE STA SUMMARY =====
Path Type: max
Path Type: max
worst slack max 8.33
tns max 0.00
worst slack min 0.48
tns min 0.00
===== POST-ROUTE STA COMPLETED =====
sphoorti@sphoorti-inspiron153530:/opt/asic-learning/countermod
10$
```